



Power Electronics

Detailed Solutions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Power Electronics (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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“We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you.”

— Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

- 3 Unit (chapter) number — Unit III of this book
- 24 GATE exam year — 2024 (two digits; 98 = 1998)
- 7 Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:



Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at pyq.prepfusion.in.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*					
Stream: EE	Questions: 45	MCQ: 30	MSQ: 3	NAT: 12	
1M: 20	2M: 22	Descriptive: 3	Avg Weightage ~4M	Range: 2M(2015)–6M(2021)	

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

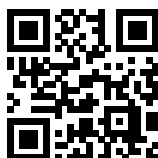
A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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Contents

1. DC to DC Converters-Choppers 1



SOLUTIONS



DC to DC Converters-Choppers

Topics

1. Buck Converter or Step Down Chopper with R/RL/RLE Load
2. Step Up Chopper or Boost Converter
3. Buck-Boost Converter
4. Fourier Series Analysis and Design of DC-DC Converters
5. Commutation Circuits

Unit Snapshot*

Stream: EE	Questions: 51	MCQ: 27	MSQ: 0	NAT: 24
1M: 14	2M: 33	Descriptive: 4	Avg Weightage ~3M	Range: 1M(2023)–6M(2013)

*See preface for how Avg Weightage and Range are calculated.

Q3.26.1 MCQ 2026 2M Ans: C ● ● ○

Given that the inductor current rises for $10\ \mu\text{s}$ and falls for $10\ \mu\text{s}$ over a complete switching cycle, the duty cycle (D) of the converter is calculated as:

$$D = \frac{T_{\text{on}}}{T_{\text{on}} + T_{\text{off}}} = \frac{10\ \mu\text{s}}{10\ \mu\text{s} + 10\ \mu\text{s}} = 0.5$$

For an ideal boost converter operating in continuous conduction mode (CCM), the relationship between input voltage (V_{in}) and output voltage (V_{out}) is expressed as:

$$V_{\text{out}} = \frac{V_{\text{in}}}{1 - D} = \frac{V_{\text{in}}}{1 - 0.5} = 2V_{\text{in}}$$

Average inductor current is 3 A

$$I_{\text{out}} = \frac{V_{\text{out}}}{10}$$

Applying the input-to-output power balance equation ($P_{\text{in}} = P_{\text{out}}$) for an ideal lossless converter:

$$V_{\text{in}} \times I_{L(\text{avg})} = \frac{V_{\text{out}}^2}{10}$$

Substituting $I_{L(\text{avg})} = 3\ \text{A}$ and $V_{\text{out}} = 2V_{\text{in}}$ into the power balance equation:

$$V_{\text{in}} \times 3 = \frac{(2V_{\text{in}})^2}{10}$$

$$3V_{\text{in}} = \frac{4V_{\text{in}}^2}{10}$$

Assuming $V_{\text{in}} \neq 0$, we can divide both sides by V_{in} :

$$3 = \frac{4V_{\text{in}}}{10}$$

$$4V_{\text{in}} = 30 \implies V_{\text{in}} = 7.5\ \text{V}$$

7.5

Key Points

- **Boost Converter Voltage Ratio:** In critical conduction mode, a boost converter steps up voltage according to the conversion ratio $V_{\text{out}} = \frac{V_{\text{in}}}{1-D}$.
- **Power Conservation Principle:** For any ideal DC-DC converter, average input power equals average output power ($V_{\text{in}}I_{\text{in}} = V_{\text{out}}I_{\text{out}}$).

Q3.25.1 NAT 2025 1M Ans: 90 ● ○ ○

For a Boost Converter operating under continuous conduction mode (CCM), the relationship between the average output voltage V_0 and the input source voltage V_s is given as:

$$V_0 = \frac{V_s}{1 - \alpha}$$

Assuming an ideal lossless converter, the input power equals the output power ($P_s = P_0$), which gives the relation between voltage and current ratios:

$$\frac{V_0}{V_s} = \frac{I_s}{I_0} = \frac{1}{1 - \alpha}$$

From this, the average output current I_0 can be expressed in terms of the source current I_s and the duty cycle α :

$$I_0 = (1 - \alpha)I_s$$

Given the operating parameters $\alpha = 0.4$ and $I_s = 10$ A:

$$I_0 = (1 - 0.4) \times 10 = 6 \text{ A}$$

Given that the output voltage $V_0 = 15$ V, the total output power P_0 delivered to the load is:

$$P_0 = V_0 I_0 = 15 \times 6 = 90 \text{ W}$$

90 W

Q3.25.2

NAT

2025

2M

Ans: 1.8

For a DC-DC Buck Converter, the given parameters yield the following step-by-step analysis:

The peak-to-peak inductor ripple current ΔI_L and capacitor ripple current ΔI_C are given as:

$$\Delta I_L = \Delta I_C = 0.2 \text{ A}$$

The switching frequency f is calculated from the total time period $T = 50 \times 10^{-6}$ s:

$$f = \frac{1}{T} = \frac{1}{50 \times 10^{-6}} = 20 \text{ kHz}$$

The duty cycle α is found using the turn-on time $T_{ON} = 30 \mu\text{s}$ and total time period $T = 50 \mu\text{s}$:

$$\alpha = \frac{T_{ON}}{T} = \frac{30}{50} = 0.6$$

The fundamental formula for the peak-to-peak inductor ripple current ΔI_L in a buck converter is:

$$\Delta I_L = \frac{\alpha(1 - \alpha)V_s}{fL}$$

Rearranging the equation to solve for the filter inductance L and substituting the given values ($V_s = 30$ V):

$$L = \frac{\alpha(1 - \alpha)V_s}{\Delta I_L \times f} = \frac{0.6(1 - 0.6) \times 30}{0.2 \times 20 \times 10^3}$$

$$L = 1.8 \text{ mH}$$

1.8 mH

Q3.24.1

NAT

2024

1M

Ans: 50

Voltage Commutated Chopper

$$t_{cm} = \frac{C}{I_0} V_c$$

$$= \frac{10 \cdot 10^{-6}}{10} \cdot 50$$

$$= 50 \mu\text{s}$$

50

Q3.24.2 NAT 2024 2M Ans: 13

For a DC-DC Boost Converter operating under continuous conduction mode (CCM), the average output voltage V_0 is related to the input source voltage V_s and the duty cycle α by:

$$V_0 = \frac{V_s}{1 - \alpha}$$

Given parameters: Input voltage, $V_s = 20$ V; Output voltage, $V_0 = 40$ V; Load resistance, $R = 10$ Ω ;

Switching frequency, $f = 500$ Hz

Substituting the voltages to determine the operating duty cycle α :

$$40 = \frac{20}{1 - \alpha} \implies 1 - \alpha = \frac{20}{40} = 0.5$$

$$\alpha = 0.5$$

The average output current I_0 is:

$$I_0 = \frac{V_0}{R} = \frac{40}{10} = 4 \text{ A}$$

The average inductor current I_L for a boost converter equals the average input source current, which can be computed as:

$$I_L = \frac{I_0}{1 - \alpha} = \frac{4}{1 - 0.5} = 8 \text{ A}$$

The peak-to-peak inductor ripple current ΔI_L is given by the standard formulation:

$$\Delta I_L = \frac{\alpha V_s}{fL}$$

$$\Delta I_L = \frac{0.5 \times 20}{500 \times 2 \times 10^{-3}} = \frac{10}{1} = 10 \text{ A}$$

The peak inductor current $I_{\max}$ represents the maximum value reached by the inductor current waveform:

$$I_{\max} = (i_L)_{\text{peak}} = I_L + \frac{\Delta I_L}{2}$$

Substituting the average value and the calculated ripple current:

$$I_{\max} = 8 + \frac{10}{2} = 8 + 5 = 13 \text{ A}$$

13

Key Points

- The average inductor current in a boost converter is higher than the output current because the inductor conducts during both the switch-on and switch-off subintervals.

Q3.23.1 MCQ 2023 1M Ans: B

Let the voltage across the 5 A current source be V_o .

During the period from $t = 0$ to $t = 3$, the switch S_1 conducts, and the voltage across the source equals the supply voltage $V_s = 20$ V. From $t = 3$ to $t = 10$, the freewheeling diode D_2 conducts, short-circuiting the voltage across this parallel path, making $v_o = 0$ V.

The corresponding waveform of V_o over one complete cycle is represented below:

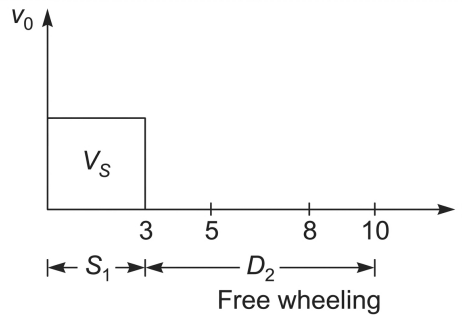


Fig. Solution Q3.23.1

The average output voltage $V_{o(\text{avg})}$ over the total time period $T = 10$ is calculated by finding the area under the curve divided by the base period:

$$V_{o(\text{avg})} = \frac{1}{T} \int_0^T v_o dt = \frac{V_s \times t_{\text{on}}}{T}$$

Substituting the given values:

$$V_{o(\text{avg})} = \frac{20 \times 3}{10} = 6 \text{ V}$$

(B) 6

Q3.22.1

NAT

2022

2M

Ans: 168



The average inductor current I_L is calculated as the mean of its maximum and minimum values:

$$I_L = \frac{I_{\text{max}} + I_{\text{min}}}{2} = \frac{16 + 12}{2} = 14 \text{ A}$$

For the given converter configuration, the average output current I_o is related to the inductor current and the duty cycle parameter α by:

$$I_o = I_L(1 - \alpha)$$

Substituting the given operational parameters $\alpha = \frac{2}{5}$:

$$I_o = 14 \times \left(1 - \frac{2}{5}\right) = 14 \times \left(\frac{3}{5}\right) = 14 \times \left(\frac{30}{50}\right) = 8.4 \text{ A}$$

The peak-to-peak output voltage ripple ΔV_o is equal to the capacitor voltage ripple ΔV_c , which is expressed as:

$$\Delta V_o = \Delta V_c = \frac{\alpha \cdot I_o}{f \cdot C}$$

Given that the permissible ripple voltage $\Delta V_o = 1 \text{ V}$ and the operating frequency $f = \frac{1}{50 \times 10^{-6}} \text{ Hz}$ (derived from the period representation):

$$1 = \frac{\left(\frac{2}{5}\right) \times 8.4}{\left(\frac{1}{50 \times 10^{-6}}\right) \cdot C}$$

Solving for the filter capacitance C :

$$C = \frac{\left(\frac{2}{5}\right) \times 8.4}{\frac{1}{50 \times 10^{-6}}} = 168 \times 10^{-6} \text{ F} = 168 \mu\text{F}$$

168

Key Points

- Average Inductor Current: For linear ripple approximations, the average current through an inductor is simply the arithmetic mean of its peak bounds: $I_{L(\text{avg})} = \frac{I_{\text{max}} + I_{\text{min}}}{2}$.
- Output Ripple Voltage: The voltage ripple across the filtering capacitor depends directly on the charge transferred during the switching intervals, given by $\Delta V_c = \frac{\Delta Q}{C}$.

Q3.21.1
NAT
2021
2M
Ans: 1.6


For a boost converter in CCM:

$$V_0 = \frac{V_S}{1 - \alpha} = \frac{15}{1 - 0.6} = 37.5 \text{ V}$$

$$I_0 = \frac{V_0}{R} = \frac{37.5}{10} = 3.75 \text{ A}$$

Using ideal power balance:

$$I_S = \frac{I_0}{1 - \alpha} = \frac{3.75}{0.4} = 9.375 \text{ A}$$

Verifying CCM operation

For the boost converter, the inductor ripple current is:

$$\Delta I_L = \frac{\alpha V_S}{fL} = \frac{0.6 \times 15}{25 \times 10^3 \times 1 \times 10^{-3}} = 0.36 \text{ A}$$

Minimum inductor current:

$$I_{L,\text{min}} = I_S - \frac{\Delta I_L}{2} = 9.375 - 0.18 = 9.195 \text{ A} > 0$$

Hence, the converter operates in CCM.

Therefore, input resistance is:

$$R_{\text{in}} = \frac{V_S}{I_S} = \frac{15}{9.375} = 1.6 \Omega$$

1.6

Key Points

- For boost converter CCM: $V_0 = \frac{V_S}{1 - \alpha}$.
- Equivalent input resistance: $R_{\text{in}} = (1 - \alpha)^2 R$.

Q3.21.2
NAT
2021
2M
Ans: 24


Given switching parameters:

$$\alpha = 0.75, \quad f = 25 \text{ kHz}$$

Calculation of Output Voltage (V_0) Assuming continuous conduction mode operation:

$$V_0 = \frac{\alpha V_s}{1 - \alpha} = \frac{0.75 \times 20}{1 - 0.75} = 60 \text{ V}$$

Calculation of Currents and Conduction Mode Verification The average output current (I_0) for a load resistance $R = 10\ \Omega$ is:

$$I_0 = \frac{V_0}{R} = \frac{60}{10} = 6\text{ A}$$

The average inductor current (I_L) is:

$$I_L = \frac{I_0}{1 - \alpha} = \frac{6}{1 - 0.75} = 24\text{ A}$$

The peak-to-peak inductor ripple current (ΔI_L) is calculated as:

$$\Delta I_L = \frac{\alpha V_s}{fL} = \frac{0.75 \times 20}{25 \times 10^3 \times (1 \times 10^{-3})} = 0.6\text{ A}$$

Now, we evaluate the minimum inductor current ($I_{L,\min}$) to verify our continuous conduction assumption:

$$I_{L,\min} = I_L - \frac{\Delta I_L}{2} = 24 - \frac{0.6}{2} = 24 - 0.3$$

$$I_{L,\min} = 23.7\text{ A}$$

Since $I_{L,\min} = 23.7\text{ A} > 0$, the continuous conduction assumption is correct.

Therefore, the average inductor current is:

24

Q3.20.1

NAT

2020

2M

Ans: 5

● ● ○

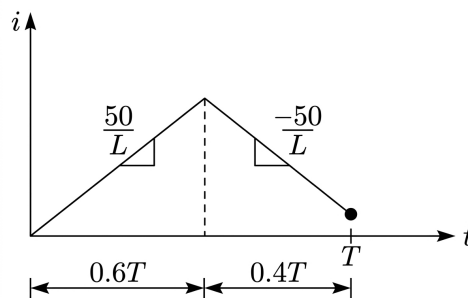


Fig. Solution Q3.20.1

Given:

- Duty cycle, $D = 0.6$
- Inductance, $L = 10\text{ mH} = 10 \times 10^{-3}\text{ H}$
- Switching frequency, $f = 10\text{ kHz}$

The effective current rise per cycle is:

$$(2D - 1)T = (2 \times 0.6 - 1)T = 0.2T$$

For 10 cycles, total effective time:

$$t = 10(0.2T) = 2T$$

Using

$$i = \frac{V}{L}t$$

with $V = 50\text{ V}$:

$$i = \frac{50}{L}(2T) = \frac{100}{Lf}$$

$$i = \frac{100}{(10 \times 10^{-3})(10 \times 10^3)} = 1 \text{ A}$$

Energy stored in the inductor:

$$E = \frac{1}{2}Li^2$$

$$E = \frac{1}{2}(10 \times 10^{-3})(1)^2 = 5 \times 10^{-3} \text{ J} = 5 \text{ mJ}$$

5

Key Points

- Inductor current changes according to $\Delta i = \frac{V}{L}\Delta t$.
- Stored energy in an inductor is $E = \frac{1}{2}Li^2$.

Q3.19.1

MCQ

2019

2M

Ans: A or C

● ● ○

Given that:

Supply voltage, $V_s = 48 \text{ V}$

Load resistance, $R_L = 24 \Omega$

Switch frequency, $f_s = 250 \text{ Hz}$

On time of switch (T_{ON}) = 1 ms

$$\text{Time period, } T = \frac{1}{f_s} = \frac{1}{250} = 4 \text{ ms}$$

$$\text{Duty cycle, } \alpha = \frac{T_{ON}}{T} = \frac{1 \text{ ms}}{4 \text{ ms}} = 0.25$$

$$\begin{aligned} \text{Load power} &= \frac{(V_0)^2}{R} = \frac{(\alpha V_s)^2}{24} \\ &= \frac{(0.25 \times 48)^2}{24} = \frac{12^2}{24} \\ &= 6 \text{ Watt} \end{aligned}$$

A

Other solution:

$$\begin{aligned} \text{Load power } (P_{\text{rms}}) &= \frac{V_{\text{rms}}^2}{R} = \frac{(\sqrt{\alpha} V_s)^2}{R} \\ &= \frac{0.25 \times 48^2}{24} = 24 \text{ W} \end{aligned}$$

C

Q3.19.2 NAT 2019 2M Ans: 24

Given: $P_0 = 120 \text{ W}$, $V_s = 24 \text{ V}$, $V_0 = 48 \text{ V}$

$$V_0 = \frac{V_s}{(1-D)}$$

$$1-D = \frac{24}{48}$$

$$D = 0.5$$

$$P_0 = V_0 I_0 = 120$$

$$I_0 = \frac{120}{48} = 2.5 \text{ A}$$

$$V_s I_s = V_0 I_0$$

$$I_s = \frac{120}{24} = 5 \text{ A}$$

At boundary of continuous and discontinuous,

$$I_L = I_s = \frac{\Delta I_L}{2}$$

$$\Delta I_L = \frac{DV_s}{fL_c} = 2 \times 5$$

$$L_c = \frac{0.5 \times 24}{50 \times 10^3 \times 10} = 24 \mu\text{H}$$

24

Q3.18.1 NAT 2018 2M Ans: 12.50

The duty cycle of each buck converter is

$$D = \frac{T_{\text{ON}}}{T} = \frac{0.5}{1} = 0.5$$

The output voltage of a buck converter is

$$V_o = DV_s = 0.5 \times 100 = 50 \text{ V}$$

The load current is

$$I_o = \frac{V_o}{R} = \frac{50}{1} = 50 \text{ A}$$

Hence, the output power is

$$P_o = V_o I_o = 50 \times 50 = 2500 \text{ W}$$

Assuming ideal converters,

$$V_s I_s(\text{avg}) = V_o I_o$$

Therefore,

$$I_s(\text{avg}) = \frac{2500}{100} = 25 \text{ A}$$

Since both converters share the load equally,

$$i_{S1}(\text{avg}) = \frac{25}{2} = 12.5 \text{ A}$$

Hence,

12.50

Q3.18.2 NAT 2018 2M Ans: 12.00 ●●●

Assume the converter operates in continuous conduction mode (CCM).

For a boost converter,

$$V_o = \frac{V_s}{1-D} = \frac{50}{1-0.4} = 83.33 \text{ V}$$

Since

$$83.33 \text{ V} < 150 \text{ V},$$

continuous conduction is not possible.

Therefore, the converter operates in discontinuous conduction mode (DCM).

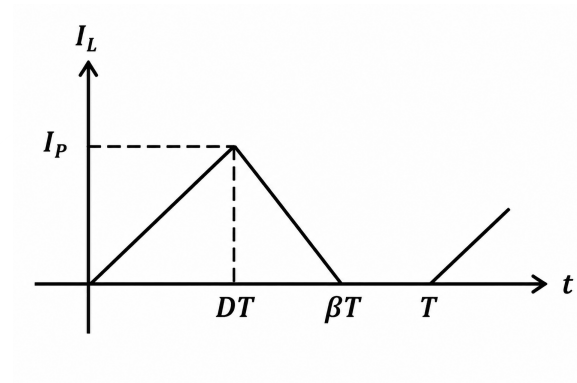


Fig. Solution Q3.18.2

For a boost converter operating in DCM,

$$V_o = \frac{\beta}{\beta-D} V_s$$

where βT is the total conduction interval of the inductor current.

Substituting,

$$150 = \frac{\beta}{\beta-0.4} \times 50$$

$$3 = \frac{\beta}{\beta-0.4}$$

$$3\beta - 1.2 = \beta$$

$$\boxed{\beta = 0.6}$$

From $0 < t < DT$, the switch is ON and

$$V_L = V_s = 50 \text{ V}$$

Using

$$V_L = L \frac{di_L}{dt},$$

$$50 = L \frac{I_p}{DT}$$

Given,

$$L = 5 \text{ mH}, \quad D = 0.4, \quad f = 5 \text{ kHz}, \quad T = \frac{1}{f} = \frac{1}{5 \times 10^3}$$

Hence,

$$I_p = \frac{50 \times 0.4}{5 \times 10^{-3} \times 5 \times 10^3} = 0.8 \text{ A}$$

The energy transferred to battery B_2 during each switching cycle is

$$E = V_o I_{\text{avg}} T$$

The average diode current over the entire switching period T is

$$I_{\text{avg}} = \frac{1}{2} I_p (\beta - D)$$

Therefore,

$$\begin{aligned} E &= 150 \times \frac{1}{2} \times 0.8 \times (\beta - D) \times T \\ &= 150 \times \frac{1}{2} \times 0.8 \times (0.6 - 0.4) \times T = 12T \end{aligned}$$

Hence, the power transferred is

$$P = \frac{E}{T} = 12 \text{ W}$$

Hence,

12.00

Q3.17.1 MCQ 2017 2M Ans: A ● ○ ○

The given relationship for the output voltage V_o of a Buck-Boost converter is:

$$V_o = \frac{\alpha V_s}{1 - \alpha}$$

When source voltage $V_s = 32 \text{ V}$ and output voltage $V_o = 48 \text{ V}$:

$$48 = \frac{\alpha \cdot 32}{1 - \alpha}$$

$$48(1 - \alpha) = 32\alpha$$

$$48 = 80\alpha$$

$$\alpha = \frac{48}{80} = \frac{3}{5}$$

Thus,

$$\alpha \leq \frac{3}{5}$$

When source voltage $V_s = 72 \text{ V}$ and output voltage $V_o = 48 \text{ V}$:

$$48 = \frac{\alpha \cdot 72}{1 - \alpha}$$

$$48(1 - \alpha) = 72\alpha$$

$$48 = 120\alpha$$

$$\alpha = \frac{48}{120} = \frac{2}{5}$$

Thus,

$$\alpha \geq \frac{2}{5}$$

Combining both cases,

$$\frac{2}{5} \leq \alpha \leq \frac{3}{5}$$

$\frac{2}{5} \leq \alpha \leq \frac{3}{5}$

Key Points

- For a Buck-Boost converter, $V_o = \frac{\alpha V_s}{1 - \alpha}$.
- The duty cycle α must satisfy both source voltage limits.

Q3.16.1

NAT

2016

1M

Ans: 0.4

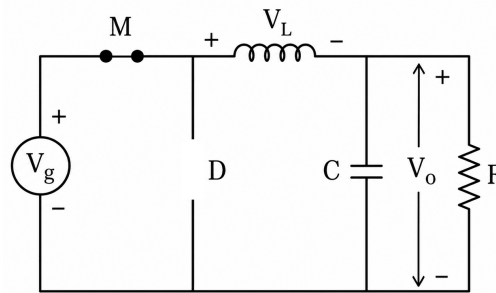


Fig. Solution Q3.16.1

During the ON interval, the inductor voltage is

$$V_L(\text{ON}) = V_g - V_o = 30 \text{ V}$$

Hence,

$$V_g - V_o = 30 \quad \dots (1)$$

During the OFF interval, the inductor voltage is

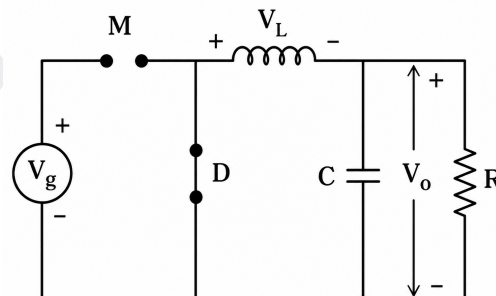


Fig. Solution Q3.16.1

$$V_L(\text{OFF}) = -V_o = -20 \text{ V}$$

Therefore,

$$V_o = 20 \text{ V} \quad \dots (2)$$

Substituting (2) into (1),

$$V_g - 20 = 30$$

$$V_g = 50 \text{ V}$$

For an ideal buck converter,

$$D = \frac{V_o}{V_g}$$

Hence,

$$D = \frac{20}{50} = 0.4$$

Therefore,

$$\boxed{0.4}$$

Q3.15.1

NAT

2015

1M

Ans: 1



For a step-down chopper,

$$V_o = \alpha V_s$$

Also,

$$V_o = I_o R + E$$

Hence,

$$I_o = \frac{V_o - E}{R} = \frac{\alpha V_s - E}{R}$$

Substituting the given values,

$$I_o = \frac{(0.4)(20) - 5}{3} = \frac{8 - 5}{3} = 1 \text{ A}$$

$$\boxed{1}$$

Q3.15.2

NAT

2015

2M

Ans: 7



Given the switching operational details:

- Switch S_1 is turned ON for 0.2 ms
- Switch S_2 is turned ON for 0.3 ms
- Switching cycle total time period (T) = 0.5 ms

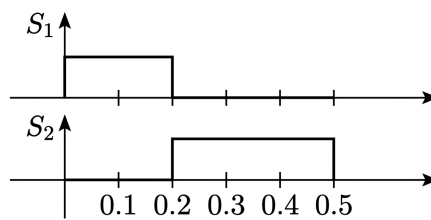


Fig. Solution Q3.15.2

From the given timing diagrams and output voltage waveform, the instantaneous output voltage $V_o(t)$ across the load resistor R_L within one complete cycle is:

$$V_o(t) = \begin{cases} 10 \text{ V}, & 0 < t \leq 0.2 \text{ ms} \\ 5 \text{ V}, & 0.2 \text{ ms} < t \leq 0.5 \text{ ms} \end{cases}$$

The average output voltage V_o over the total switching period $T = 0.5 \text{ ms} = 0.5 \times 10^{-3} \text{ s}$ is calculated using the integration method:

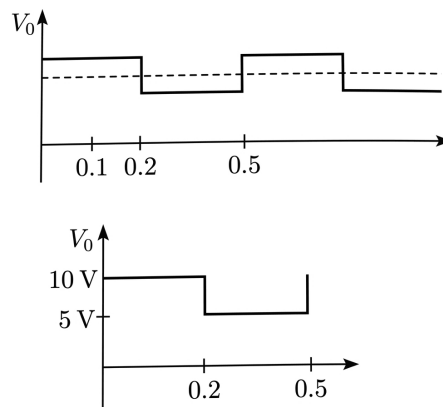


Fig. Solution Q3.15.2

$$V_o = \frac{1}{0.5 \times 10^{-3}} \left[\int_0^{0.2 \times 10^{-3}} 10 \, dt + \int_{0.2 \times 10^{-3}}^{0.5 \times 10^{-3}} 5 \, dt \right]$$

Evaluating the individual integration segments:

$$V_o = \frac{1}{0.5 \times 10^{-3}} [10(0.2 \times 10^{-3} - 0) + 5(0.5 \times 10^{-3} - 0.2 \times 10^{-3})]$$

$$V_o = \frac{1}{0.5 \times 10^{-3}} [10(0.2 \times 10^{-3}) + 5(0.3 \times 10^{-3})]$$

$$V_o = \frac{1}{0.5 \times 10^{-3}} [2 \times 10^{-3} + 1.5 \times 10^{-3}]$$

$$V_o = \frac{3.5 \times 10^{-3}}{0.5 \times 10^{-3}} = 7 \, \text{V}$$

7

Key Points

- For multi-level rectangular waveforms, the average value can be simplified to the sum of individual areas divided by the total time period:

$$V_{\text{avg}} = \frac{V_1 t_1 + V_2 t_2}{T}$$

Mistakes to be avoided

- Unit Conversion Error:** Remember to match the time units. Since the intervals are given in milliseconds (ms), ensure that either all values are sustained in ms or completely converted to seconds (10^{-3} s) to prevent decimal miscalculations.

Q3.15.3

MCQ

2015

2M

Ans: A

☒ ☐ ☐

The average voltage across the inductor in steady-state is zero ($V_L = 0$) due to the inductor volt-second balance principle.

Since the given circuit is a boost converter, the average voltage across the output capacitor is equal to the average voltage across the load. Therefore, the relationship between the output voltage V_o and the input DC voltage V_{dc} is given by:

$$V_o = \left(\frac{1}{1 - \delta} \right) V_{dc}$$

where δ represents the duty cycle of the converter.

$$(A) V_L = 0 \quad \text{and} \quad V_C = \frac{1}{1-\delta} V_{dc}$$

Q3.15.4
NAT
2015
2M
Ans: 2500


At critical conduction,

$$I_{L,avg} = \frac{D(1-D)V_s}{2fL}$$

Also,

$$I_{O,avg} = \frac{V_{O,avg}}{R}$$

Since

$$I_{L,avg} = I_{O,avg}$$

$$\frac{V_{O,avg}}{R} = \frac{D(1-D)V_s}{2fL}$$

Substituting,

$$\frac{36}{R} = \frac{60 \times 0.4 \times 0.6}{2 \times 100 \times 10^3 \times 5 \times 10^{-3}} = 0.0144$$

$$R = \frac{36}{0.0144} = 2500$$

2500

Q3.15.5
NAT
2015
2M
Ans: 0.75

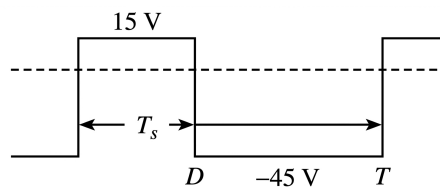
Method 1


Fig. Solution Q3.15.5

From the given inductor voltage waveform, we can determine the operation of the converter during different switching intervals.

During the interval $0 \leq t \leq T_{on}$:

$$V_s = V_L = 15 \text{ V}$$

During the interval $T_{on} \leq t \leq T$:

$$V_L = -(V_0 - V_s) = -45 \text{ V}$$

Given that the input voltage $V_s = 15 \text{ V}$:

$$-(V_0 - 15) = -45 \text{ V}$$

$$V_0 = 60 \text{ V}$$

Hence, it is a boost regulator. The relationship between output and input voltage is given by:

$$V_0 = \left(\frac{1}{1-\alpha} \right) V_s$$

Substituting the values:

$$60 = \left(\frac{1}{1 - \alpha} \right) 15$$

$$1 - \alpha = \frac{15}{60} = 0.25$$

$$\alpha = 0.75$$

Method 2

Since the inductor current is always continuous, we can apply the inductor volt-second balance equation over one complete switching period T :

$$\int_0^T V_L dt = 0$$

Let D be the duty cycle ($D = \alpha$). From the waveform:

$$15 \cdot DT + (-45) \cdot (1 - D)T = 0$$

$$15D = 45(1 - D)$$

$$60D = 45$$

$$D = \frac{45}{60} = \frac{3}{4} = 0.75$$

Therefore, the duty cycle of the regulator is:

$$0.75$$

Key Points

- **Volt-Second Balance Principle:** For an inductor operating in steady-state continuous conduction mode, the net volt-seconds applied to the inductor over a complete switching period must be zero.

Mistakes to be avoided

- Ensure that the negative sign for the inductor voltage during the OFF period is properly accounted for when setting up the volt-second balance equation.

Q3.14.1

NAT

2014

2M

Ans: 2.5



The circuit shown is a Buck chopper. The average output voltage V_0 is related to the source voltage V_s and the switching intervals by the formula:

$$V_0 = V_s \frac{T_{ON}}{T}$$

Substituting the given values into the equation:

$$V_0 = 100 \times \frac{0.05 \times 10^{-3}}{0.1 \times 10^{-3}} = 50 \text{ V}$$

The peak-to-peak ripple in the inductor current (ΔI) is given by:

$$\Delta I = \frac{V_0 T_{OFF}}{L}$$

Substituting the calculated output voltage and given system parameters:

$$= 50 \times \frac{0.05 \times 10^{-3}}{1 \times 10^{-3}}$$

$$\Delta I = 2.5 \text{ A}$$

Therefore, the peak-to-peak ripple current is:

$$\boxed{2.5}$$

Q3.14.2 NAT 2014 1M Ans: 31.2 ● ○ ○

Given a step-up (boost) chopper with the following parameters: Output voltage, $V_0 = 400 \text{ V}$ Source voltage, $V_s = 250 \text{ V}$ Turn-off time, $T_{\text{OFF}} = 20 \times 10^{-6} \text{ sec}$

The relationship between the output voltage and source voltage for a step-up chopper is given by the formula:

$$V_0 = V_s \times \frac{T}{T_{\text{OFF}}}$$

Substituting the given values into the equation to find the total chopping period T :

$$400 = 250 \times \frac{T}{20 \times 10^{-6}}$$

Solving for T :

$$T = \frac{400 \times 20 \times 10^{-6}}{250} = 3.2 \times 10^{-5} \text{ sec}$$

The chopping frequency f is the reciprocal of the total period T :

$$f = \frac{1}{T} = \frac{1}{3.2 \times 10^{-5}}$$

$$f = 31250 \text{ Hz} = 31.25 \text{ kHz}$$

Therefore, the chopping frequency is:

$$\boxed{31.25}$$

Q3.13.1 MCQ 2013 2M Ans: D ● ○ ○

The armature loop equation for a DC motor under steady-state operating conditions is defined as:

$$V_a = E_b + I_a R_a$$

Initially, substituting the known operating parameter values ($V_a = 150 \text{ V}$, $I_a = 20 \text{ A}$, and $R_a = 1 \Omega$) into the voltage relation:

$$150 = E_b + 20 \times 1$$

$$E_b = 130 \text{ V}$$

Since the electromagnetic torque developed by a DC motor is directly proportional to its armature current ($T \propto I_a$) under constant field flux conditions, operating at half the rated torque reduces the required current by half:

$$I_a = 10 \text{ A}$$

The newly required terminal voltage V_a to sustain operation at this torque level while keeping the speed (and hence the back EMF, E_b) constant is:

$$V_a = 130 + 10 \times 1 = 140 \text{ V}$$

Assuming a chopping circuit configures the variable voltage supply with a total source potential of 200 V , the required duty cycle D is calculated as:

$$D = \frac{140}{200} = 0.7$$

Therefore, the duty cycle of the converter is:

$$\boxed{(D) 0.7}$$

Key Points

- **DC Motor Voltage Equation:** The applied terminal voltage matches the sum of the internally generated back EMF and the internal resistance voltage drop ($V_a = E_b + I_a R_a$).
- **Torque-Current Dependency:** Under constant field excitation, the developed mechanical torque varies directly with the current magnitude flowing through the armature ($T \propto I_a$).

Mistakes to be avoided

- Make sure not to change the back EMF value E_b unless a change in operating speed or magnetic flux is explicitly specified in the problem statement.

Q3.13.2

MCQ

2013

2M

Ans: B



In the given circuit, it is a boost converter. The inductor is directly connected to the source, so the average source current equals the average inductor current:

$$I_{S(\text{avg})} = I_{L(\text{avg})}$$

The average output voltage $V_{o(\text{avg})}$ for a boost converter is given by:

$$V_{o(\text{avg})} = \frac{V_S}{1-D} = \frac{12}{1-0.4} = 20 \text{ V}$$

The average output current $I_{o(\text{avg})}$ is:

$$I_{o(\text{avg})} = \frac{V_{o(\text{avg})}}{R} = \frac{20}{20} = 1 \text{ A}$$

During the interval $DT < t < T$ (when the switch is off), the diode conducts, and the output current is equal to the inductor current:

$$I_o = I_L$$

Taking the average over the complete switching cycle:

$$I_{o(\text{avg})} = I_{L(\text{avg})}(1-D)$$

Therefore, the average inductor current (which is also the average source current) is:

$$I_{L(\text{avg})} = I_{S(\text{avg})} = \frac{I_{o(\text{avg})}}{1-D} = \frac{1}{1-0.4} = \frac{5}{3} \text{ A}$$

$$(B) \frac{5}{3}$$

Q3.13.3

MCQ

2013

2M

Ans: C



The peak source current is

$$I_{S(\text{peak})} = \Delta I_L$$

For the inductor ripple current,

$$\Delta I_L = \frac{V_S D}{fL}$$

Substituting,

$$\Delta I_L = \frac{12 \times 0.4}{250 \times 0.1} = 0.192 \text{ A}$$

$$(C) 0.192$$

Q3.12.1 MCQ 2012 2M Ans: C ☒ ☐ ☐

Peak inductor current is

$$I_{L(\text{peak})} = I_o + \frac{\Delta I_L}{2}$$

Substituting,

$$I_{L(\text{peak})} = 5 + \frac{1.6}{2} = 5.8 \text{ A}$$

(C) 5.8

Q3.11.1 MCQ 2011 2M Ans: A ☒ ☒ ☐

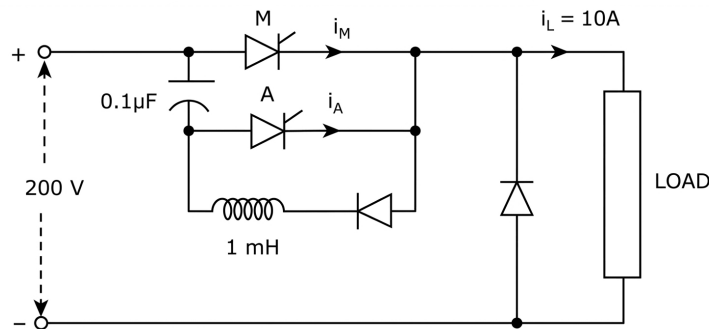


Fig. Solution Q3.11.1

The current through the main thyristor during Mode-1 of a voltage commutated chopper is

$$i_M = V_s \sqrt{\frac{C}{L}} \sin \omega t + I_o$$

Hence, the peak current through the main thyristor is

$$i_{M(\text{peak})} = V_s \sqrt{\frac{C}{L}} + I_o$$

Given,

$$V_s = 200 \text{ V}, \quad C = 0.1 \mu\text{F}, \quad L = 1 \text{ mH}, \quad I_o = 10 \text{ A}$$

Substituting,

$$\begin{aligned} i_{M(\text{peak})} &= 200 \sqrt{\frac{0.1 \times 10^{-6}}{1 \times 10^{-3}}} + 10 \\ &= 200(0.01) + 10 = 2 + 10 = 12 \text{ A} \end{aligned}$$

When the main thyristor is turned OFF, the auxiliary thyristor carries only the load current. Therefore,

$$i_{A(\text{peak})} = I_o = 10 \text{ A}$$

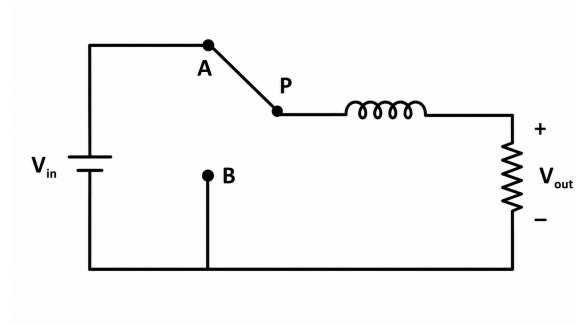
Hence, the correct answer is

$$(A) i_{M_{\text{max}}} = 12 \text{ A and } i_{A_{\text{max}}} = 10 \text{ A}$$

Q3.10.1
MCQ
2010
1M
Ans: A
☒ ☒ ☐

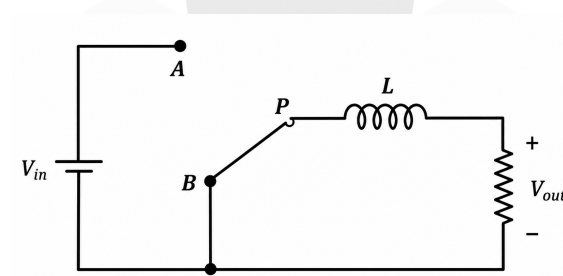
When the switch is connected to position A for a duration T_1 ,

$$V_P = V_{in}$$


Fig. Solution Q3.10.1

When the switch is connected to position B for a duration T_2 ,

$$V_P = 0$$


Fig. Solution Q3.10.1

Hence, the average voltage at point P is

$$V_P(\text{avg}) = \frac{V_{in}T_1}{T_1 + T_2} = \alpha V_{in}$$

where,

$$\alpha = \frac{T_1}{T_1 + T_2}$$

is the duty cycle.

Since the average voltage across the inductor is zero in steady state (volt-second balance),

$$V_o = V_P(\text{avg}) = \alpha V_{in}$$

As the output voltage is always less than the input voltage,

$$V_o < V_{in},$$

the converter operates as a step-down (buck) chopper.

Hence, the correct answer is

(A) Step-down chopper (buck converter)

Q3.09.1

MCQ

2009

2M

Ans: A



The given circuit is a buck converter.

The average output voltage is

$$V_o = \alpha V_s = 0.8 \times 100 = 80 \text{ V}$$

The average output current is

$$I_o = \frac{V_o}{R} = \frac{80}{8} = 10 \text{ A}$$

Since the ripple current through L_0 is negligible,

$$I_D = I_o = 10 \text{ A}$$

Initially, the commutating capacitor C_1 is charged to the supply voltage,

$$V_C = V_s = 100 \text{ V}$$

with the right plate positive.

When the auxiliary thyristor T_A is turned ON, the capacitor voltage is applied across the main thyristor T_M , producing a reverse voltage.

Hence,

$$|V_{TM}| = |V_C|$$

The maximum allowable rate of rise of the reapplied voltage is

$$\frac{dV_{TM}}{dt} = 50 \text{ V}/\mu\text{s}$$

Therefore,

$$\frac{dV_C}{dt} = 50 \text{ V}/\mu\text{s}$$

Using the capacitor current relation,

$$I_D = C_1 \frac{dV_C}{dt}$$

Substituting,

$$10 = C_1 \times 50$$

$$C_1 = \frac{10}{50} = 0.2 \mu\text{F}$$

Hence, the correct answer is

$$(A) 0.2 \mu\text{F}$$

Q3.08.1

MCQ

2008

2M

Ans: C



The given circuit is a boost converter.

For a boost converter operating in continuous conduction mode,

$$V_o = \frac{V_{in}}{1 - \alpha}$$

Given,

$$V_{in} = 20 \text{ V}, \quad \alpha = 0.5$$

Hence,

$$V_o = \frac{20}{1 - 0.5} = 40 \text{ V}$$

The average output voltage is

$$V_o = 40 \text{ V}$$

The average diode current is

$$I_D(\text{avg}) = \frac{1}{T} \int_{T_{\text{ON}}}^T i_L dt$$

Since the inductor current is continuous and constant,

$$i_L = I_L = 4 \text{ A}$$

Therefore,

$$I_D(\text{avg}) = \frac{I_L(T - T_{\text{ON}})}{T} = I_L(1 - \alpha)$$

Substituting,

$$I_D(\text{avg}) = 4(1 - 0.5) = 2 \text{ A}$$

Hence, the correct answer is

$$(C) \ 40 \text{ V}, \ 2 \text{ A}$$

Q3.07.1

MCQ

2007

2M

Ans: C



When the auxiliary thyristor Th_{Aux} is triggered at $t = 0$, the LC commutation circuit starts resonating.

The resonant angular frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Given,

$$L = 25.28 \mu\text{H}, \quad C = 10 \mu\text{F}$$

Hence, the resonant time period is

$$T = 2\pi\sqrt{LC} = 2\pi\sqrt{25.28 \times 10^{-6} \times 10 \times 10^{-6}} = 100 \mu\text{s}$$

Therefore,

$$\frac{T}{4} = 25 \mu\text{s}, \quad \frac{T}{2} = 50 \mu\text{s}$$

The peak resonant current is

$$I_p = V_s \sqrt{\frac{C}{L}} = 230 \sqrt{\frac{10 \times 10^{-6}}{25.28 \times 10^{-6}}} \approx 144.6 \text{ A}$$

Since

$$I_p > I_o = 10 \text{ A},$$

the resonant current is sufficient to commutate the main SCR.

From 0 to $50 \mu\text{s}$, the resonant current flows through the auxiliary thyristor while charging and discharging the commutating capacitor.

After $t = 50 \mu\text{s}$, the resonant current reverses and flows through the commutating diode, opposing the load current through the main SCR. During this interval, the current through the main SCR becomes zero and then reverse, thereby turning it OFF.

This reverse-current interval lasts for one-quarter of the resonant cycle, i.e.,

$$50 \mu\text{s} < t \leq 75 \mu\text{s}$$

Hence, the correct answer is

$$(C) \ 50 \mu\text{s} < t \leq 75 \mu\text{s}$$

Q3.06.1 MCQ 2006 2M Ans: B ☒ ☒ ☐

The minimum ON time of the main SCR is equal to the resonant half-cycle of the commutation circuit.

The resonant angular frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Hence, the minimum ON time is

$$T_{\min} = \frac{\pi}{\omega_0} = \pi\sqrt{LC}$$

Given,

$$L = 2 \text{ mH}, \quad C = 1 \mu\text{F}$$

Substituting,

$$\begin{aligned} T_{\min} &= \pi\sqrt{2 \times 10^{-3} \times 1 \times 10^{-6}} \\ &= 140 \mu\text{s} \end{aligned}$$

Hence, the correct answer is

$$(B) 140 \mu\text{s}$$

Q3.06.2 MCQ 2006 2M Ans: B ☒ ☒ ☐

The average output voltage of a voltage commutated chopper is

$$(V_o)_{\text{avg}} = V_s \left(\frac{T_{\text{ON}} + 2t_{\text{cm}}}{T} \right)$$

The commutation time is

$$t_{\text{cm}} = \frac{CV_s}{I_o}$$

Given,

$$C = 1 \mu\text{F}, \quad V_s = 250 \text{ V}, \quad I_o = 10 \text{ A}$$

Therefore,

$$t_{\text{cm}} = \frac{1 \times 10^{-6} \times 250}{10} = 25 \mu\text{s}$$

From Question Q3.06.1,

$$T_{\text{ON}(\min)} = 140 \mu\text{s}$$

The switching frequency is

$$f = 1 \text{ kHz}$$

Hence,

$$T = \frac{1}{f} = 1 \text{ ms}$$

For minimum average output voltage,

$$T_{\text{ON}} = 140 \mu\text{s}$$

Substituting,

$$\begin{aligned} (V_o)_{\min} &= 250 \left(\frac{140 \times 10^{-6} + 2(25 \times 10^{-6})}{1 \times 10^{-3}} \right) \\ &= 250(0.19) = 47.5 \text{ V} \end{aligned}$$

Hence, the correct answer is

$$(B) 47.5 \text{ V}$$

Q3.05.1

MCQ

2005

2M

Ans: C



The switching frequency is

$$f = 1 \text{ kHz}$$

Hence, the switching period is

$$T = \frac{1}{f} = \frac{1}{10^3} = 10^{-3} \text{ s}$$

The peak-to-peak ripple current in a buck converter operating in continuous conduction mode is

$$\Delta I_L = \frac{V_s D(1 - D)}{fL}$$

Given,

$$V_s = 100 \text{ V}, \quad D = 0.5, \quad L = 200 \text{ mH} = 200 \times 10^{-3} \text{ H}$$

Substituting,

$$\begin{aligned} \Delta I_L &= \frac{100 \times 0.5(1 - 0.5)}{10^3 \times 200 \times 10^{-3}} \\ &= \frac{100}{4 \times 10^3 \times 200 \times 10^{-3}} = 0.125 \text{ A} \end{aligned}$$

Hence, the correct answer is

(C) 0.125 A

Q3.04.1

MCQ

2004

2M

Ans: A



The given circuit is a step-down (buck) chopper.

The average output voltage is

$$V_o = \alpha V_s = 0.8 \times 100 = 80 \text{ V}$$

The average load current is

$$I_o = \frac{V_o}{R} = \frac{80}{10} = 8 \text{ A}$$

Since the load inductance is sufficiently large, the load current is ripple-free.

The diode conducts only during the OFF interval of the switch. Hence, the average diode current is

$$\begin{aligned} I_D &= \frac{1}{T} \int_{T_{ON}}^T I_o dt \\ &= \frac{T - T_{ON}}{T} I_o \end{aligned}$$

Since,

$$\alpha = \frac{T_{ON}}{T},$$

$$I_D = (1 - \alpha) I_o$$

Substituting,

$$I_D = (1 - 0.8) \times 8 = 1.6 \text{ A}$$

Hence, the correct answer is

(A) 1.6 A

Q3.04.2

MCQ

2004

2M

Ans: D



The capacitor current in the voltage commutated chopper is

$$i_C = V_s \sqrt{\frac{C}{L}} \sin \omega t$$

The current through the main switch S_1 is the sum of the load current and the capacitor current.

$$i_{S_1} = I_o + i_C$$

Hence,

$$i_{S_1} = I_o + V_s \sqrt{\frac{C}{L}} \sin \omega t$$

The peak current occurs when

$$\sin \omega t = 1.$$

Therefore,

$$i_{S_1(\text{peak})} = I_o + V_s \sqrt{\frac{C}{L}}$$

Given,

$$\begin{aligned} I_o &= 20 \text{ A}, & V_s &= 200 \text{ V}, \\ C &= 2 \mu\text{F}, & L &= 200 \mu\text{H} \end{aligned}$$

Substituting,

$$\begin{aligned} i_{S_1(\text{peak})} &= 20 + 200 \sqrt{\frac{2 \times 10^{-6}}{200 \times 10^{-6}}} \\ &= 20 + 200 \sqrt{0.01} = 20 + 20 = 40 \text{ A} \end{aligned}$$

Hence, the correct answer is

(D) 40 A

Q3.03.1

MCQ

2003

2M

Ans: A



When the chopper is ON,

$$0 \leq t \leq 200 \mu\text{s},$$

the output voltage is

$$V_o = 60 \text{ V}$$

The battery voltage is

$$V_B = 12 \text{ V}$$

Hence, the voltage across the inductor is

$$V_L = V_o - V_B = 60 - 12 = 48 \text{ V}$$

Using the inductor equation,

$$V_L = L \frac{di}{dt}$$

If the change in current during the ON interval is ΔI (which is the peak-to-peak ripple current),

$$\Delta I = \frac{V_L T_{\text{ON}}}{L}$$

Given,

$$L = 20 \text{ mH}, \quad T_{\text{ON}} = 200 \mu\text{s}$$

Substituting,

$$\Delta I = \frac{48 \times 200 \times 10^{-6}}{20 \times 10^{-3}} = 0.48 \text{ A}$$

Hence, the correct answer is

$$(A) \ 0.48 \text{ A}$$

Q3.02.1

MCQ

2002

1M

Ans: A



For a step-down (buck) chopper operating in continuous conduction mode, the average output voltage is

$$V_o = DV_s$$

where,

$$D = \frac{T_{\text{ON}}}{T}$$

is the duty cycle.

Therefore,

$$\frac{V_o}{V_s} = D$$

Hence, the correct answer is

$$(A) \ D$$

Q3.02.2

MCQ

2002

2M

Ans: C



The given circuit is a step-down (buck) converter operating in continuous conduction mode.

The average output voltage is

$$V_o = DV_s$$

The peak-to-peak inductor current ripple is

$$\Delta I_L = \frac{V_o T_{\text{OFF}}}{L} = \frac{DV_s(1-D)}{Lf}$$

At the boundary between continuous and discontinuous conduction,

$$I_{L(\text{min})} = 0$$

Hence,

$$\Delta I_L = I_{L(\text{max})} - I_{L(\text{min})} = I_{L(\text{max})}$$

Since,

$$I_{L(\text{avg})} = \frac{I_{L(\text{max})} + I_{L(\text{min})}}{2},$$

we get

$$\frac{\Delta I_L}{2} = I_{L(\text{avg})} = I_o$$

Therefore,

$$I_o = \frac{DV_s(1-D)}{2Lf}$$

Also,

$$I_o = \frac{V_o}{R} = \frac{DV_s}{R}$$

Equating the two expressions,

$$\frac{DV_s}{R} = \frac{DV_s(1-D)}{2Lf}$$

Cancelling DV_s ,

$$1-D = \frac{2Lf}{R}$$

Since,

$$f = \frac{1}{T},$$

$$1-D = \frac{2L}{RT}$$

Hence,

$$D = 1 - \frac{2L}{RT}$$

Therefore, the correct answer is

$$(C) D = 1 - \frac{2L}{RT}$$

Q3.02.3

NAT

2002

5M

Ans: *

☒ ☒ ☐

The given circuit is a buck-boost converter (step-up/step-down chopper).

(a) The output voltage of a buck-boost converter operating in continuous conduction mode is

$$\frac{V_o}{V_s} = \frac{D}{1-D}$$

Given,

$$V_s = 100 \text{ V}, \quad V_o = 500 \text{ V}$$

Hence,

$$\frac{D}{1-D} = \frac{500}{100} = 5$$

$$D = 5(1-D)$$

$$D = \frac{5}{6}$$

Since,

$$D = \frac{T_{ON}}{T}$$

and

$$T = 100 \mu\text{s},$$

$$T_{ON} = \frac{5}{6} \times 100 = 83.33 \mu\text{s}$$

(b) At the boundary of continuous and discontinuous conduction,

$$I_{\min} = 0$$

During the ON interval, the inductor voltage is

$$V_L = V_s = 100 \text{ V}$$

Using,

$$V_L = L \frac{\Delta I}{T_{ON}},$$

$$100 = L \frac{I_p - 0}{T_{ON}}$$

Given,

$$L = 100 \mu\text{H}$$

Hence,

$$I_p = \frac{100 \times 83.33 \times 10^{-6}}{100 \times 10^{-6}} = 83.33 \text{ A}$$

Hence,

$$T_{ON} = 83.33 \mu\text{s}, \quad I_p = 83.33 \text{ A}$$

Q3.02.4

NAT

2002

5M

Ans: *



(a) The capacitor voltage V_c

When the switch is ON, $0 \leq t \leq DT$, the capacitor discharges through the resistor. Hence,

$$i_C = -\frac{V_c}{R}$$

where the negative sign indicates that the capacitor is discharging.

When the switch is OFF, $DT \leq t \leq T$, the capacitor is charged by the constant current source.

Hence, $i_C = I$

Since the capacitor voltage has negligible ripple, charge balance over one switching period gives

$$\int_0^T i_C dt = 0$$

Therefore,

$$-\frac{V_c}{R}(DT) + I(T - DT) = 0$$

$$-\frac{V_c}{R}D + I(1 - D) = 0$$

Hence,

$$V_c = IR \frac{1 - D}{D}$$

(b) Average output voltage

During the ON interval,

$$0 \leq t \leq DT,$$

the diode is reverse biased.

Hence,

$$V_o = -V_c$$

During the OFF interval,

$$DT \leq t \leq T,$$

the diode conducts and short-circuits the output.

Hence,

$$V_o = 0$$

Therefore, the average output voltage is

$$V_o = \frac{1}{T} \left[\int_0^{DT} (-V_c) dt + \int_{DT}^T 0 dt \right]$$

$$V_o = -\frac{V_c DT}{T} = -DV_c$$

Substituting the value of V_c ,

$$V_o = -D \left(IR \frac{1-D}{D} \right)$$

$$V_o = -IR(1-D)$$

Hence,

$$V_c = IR \frac{1-D}{D}, \quad V_o = -IR(1-D)$$

Q3.01.1

NAT

2001

5M

Ans: *



(a) Circuit turn-off time of the main thyristor

For a voltage commutated chopper, the circuit turn-off time is

$$t_c = \frac{CV_s}{I_o}$$

Given,

$$C = 6 \mu\text{F}, \quad V_s = 100 \text{ V}, \quad I_o = 20 \text{ A}$$

Substituting,

$$t_c = \frac{6 \times 10^{-6} \times 100}{20} = 30 \times 10^{-6} \text{ s} = 30 \mu\text{s}$$

(b) Value of inductance L

The peak current through the main thyristor is

$$I_{m(\text{peak})} = V_s \sqrt{\frac{C}{L}} + I_o$$

The peak current is limited to 180% of the load current.

Hence,

$$I_{m(\text{peak})} = 1.8I_o$$

Therefore,

$$V_s \sqrt{\frac{C}{L}} + I_o = 1.8I_o$$

$$100 \sqrt{\frac{C}{L}} = 0.8 \times 20 = 16$$

Substituting $C = 6 \times 10^{-6} \text{ F}$,

$$100 \sqrt{\frac{6 \times 10^{-6}}{L}} = 16$$

$$10^4 \left(\frac{6 \times 10^{-6}}{L} \right) = 256$$

$$L = \frac{6 \times 10^{-2}}{256} = 2.34 \times 10^{-4} \text{ H} = 0.234 \text{ mH}$$

(c) Maximum instantaneous output voltage

The maximum instantaneous output voltage of a voltage commutated chopper is

$$V_{o(\max)} = 2V_s$$

Hence,

$$V_{o(\max)} = 2 \times 100 = 200 \text{ V}$$

Hence,

$$t_c = 30 \mu\text{s}, \quad L = 0.234 \text{ mH}, \quad V_{o(\max)} = 200 \text{ V}$$

Q3.00.1

MCQ

2000

2M

Ans: C



During the current rise interval t_r , the main switch is ON and the source is connected to the load.

Hence,

$$V_o = V_s$$

During the current fall interval t_f , the main switch is OFF and the load current freewheels through the diode. Therefore,

$$V_o = 0$$

After the load current decays to zero, the freewheeling diode stops conducting and only the motor back emf appears across the load.

Thus, during the interval t_0 ,

$$V_o = E_b$$

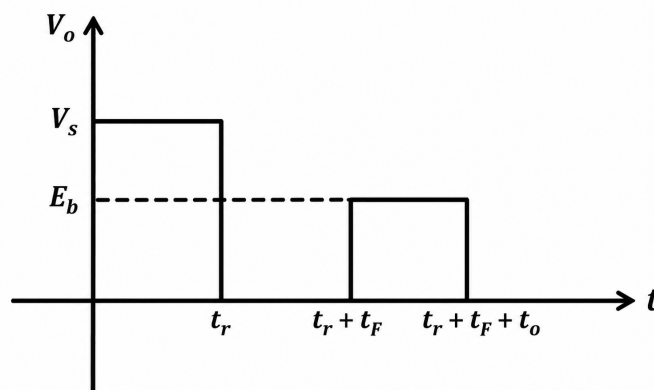


Fig. Solution Q3.00.1

The average output voltage is

$$V_{o(\text{avg})} = \frac{1}{t_r + t_f + t_0} (V_s t_r + 0 \times t_f + E_b t_0)$$

Hence,

$$V_{o(\text{avg})} = \frac{V_s t_r + E_b t_0}{t_r + t_f + t_0}$$

Therefore, the correct answer is

$$(C) \frac{V_s t_r + E_b t_0}{t_r + t_f + t_0}$$

Q3.99.1 MCQ 1999 1M Ans: B ☒ ☐ ☐

Resonant converters employ Zero Voltage Switching (ZVS) or Zero Current Switching (ZCS), so the power switches are turned ON or OFF under soft-switching conditions. This significantly reduces the overlap between device voltage and current during switching, thereby minimizing switching losses and improving converter efficiency. Hence, resonant converters are primarily used to reduce switching losses. Therefore, the correct answer is

(B) Reduce the switching losses

Q3.96.1 MCQ 1996 1M Ans: A ☒ ☐ ☐

Voltage commutation provides the best performance in a DC chopper. It offers advantages such as:

- Higher average output voltage.
- Faster and more reliable turn-off of the thyristor.
- Better overall efficiency and performance.

Hence, the correct answer is

(A) Voltage commutation

Q3.95.1 NAT 1995 1M Ans: 40 ☒ ☐ ☐

For a voltage commutated chopper, the thyristor turn-off time is

$$T_{\text{off}} = \frac{CV_s}{I_o}$$

Given,

$$C = 2 \mu\text{F}, \quad V_s = 200 \text{ V}, \quad I_o = 10 \text{ A}$$

Substituting,

$$T_{\text{off}} = \frac{2 \times 10^{-6} \times 200}{10} = 40 \times 10^{-6} \text{ s} = 40 \mu\text{s}$$

Hence,

40

Q3.95.2 MCQ 1995 1M Ans: A ☒ ☒ ☐

The average load current is

$$I_o = \frac{\alpha V_s}{R}$$

where α is the duty cycle.

Since the average load current is to remain unchanged,

$$\alpha = \text{constant}$$

The ripple in the load current is given by

$$\Delta I_L = \frac{V_s \alpha (1 - \alpha)}{fL}$$

Thus,

$$\Delta I_L \propto \frac{1}{f}$$

Hence, increasing the chopper frequency while keeping the duty cycle constant reduces the load current ripple without changing the average load current.

Therefore, the correct answer is

(A) Increase the chopper frequency keeping its duty cycle constant

Q3.93.1

MCQ

1993

1M

Ans: C

● ● ○

The ripple in the load current of a chopper-fed $R-L$ load is given by

$$\Delta I_L = \frac{V_s \alpha (1 - \alpha)}{fL}$$

where,

α = Duty cycle

Since

$$\Delta I_L \propto \alpha (1 - \alpha),$$

the maximum ripple occurs when

$$\frac{d}{d\alpha} [\alpha (1 - \alpha)] = 0$$

$$1 - 2\alpha = 0$$

$$\Rightarrow \alpha = 0.5$$

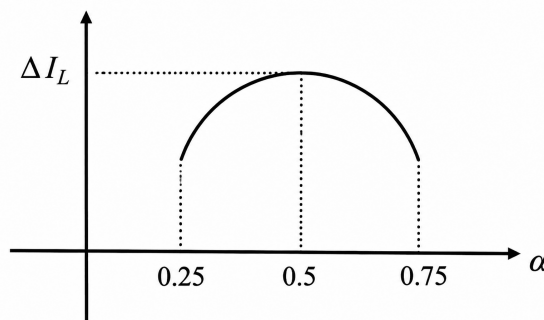
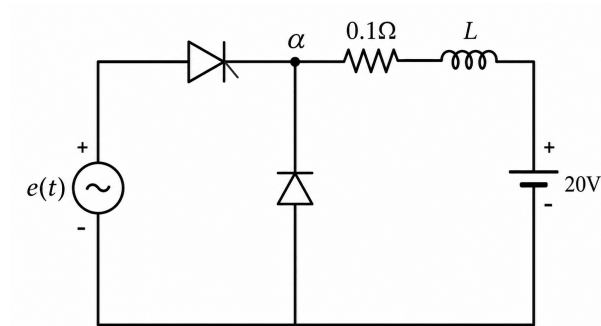


Fig. Solution Q3.93.1

Thus, as the duty cycle increases from 25% to 50%, the ripple increases. Beyond 50%, the ripple decreases. Hence, the correct answer is

(C) Increases, reaches a maximum at 50% duty ratio and then decreases

Q3.92.1
NAT
1992
1M
Ans: positive, 109.485

Fig. Solution Q3.92.1

Since the inductance is large, the load current is continuous.
The average output voltage is

$$V_o = I_o R + E \quad [\text{where, } R = 0.1 \, \Omega, \, E = 20 \, \text{V}]$$

Also,

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha)$$

Given,

$$V_m = 200\sqrt{2} \, \text{V}, \quad I_o = 100 \, \text{A},$$

Substituting,

$$\frac{200\sqrt{2}}{2\pi} (1 + \cos \alpha) = (100 \times 0.1) + 20$$

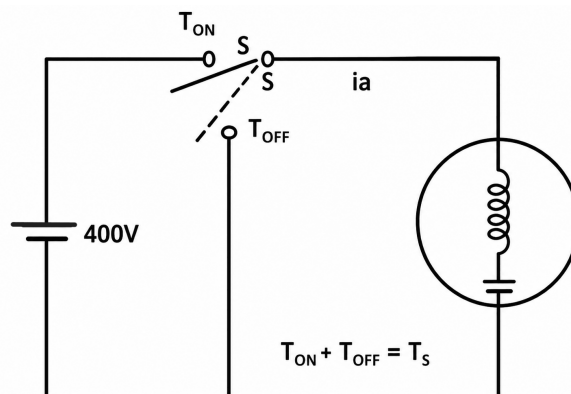
$$\Rightarrow 1 + \cos \alpha = 0.4241$$

$$\cos \alpha = -0.5759$$

$$\Rightarrow \alpha = 109.485^\circ$$

Since the thyristor conducts only during the positive half-cycle of the supply, the thyristor is triggered in the **positive** half-cycle at a delay angle of

$$109.485^\circ$$

Q3.91.1
NAT
1991
5M
Ans: *

Fig. Solution Q3.91.1

(a) The output voltage waveform of the chopper is shown below.

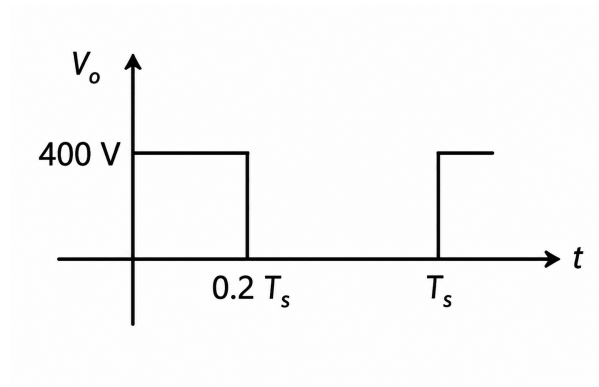


Fig. Solution Q3.91.1

During $T_{ON} = 0.2T_s$,

$$v_o(t) = 400 \text{ V}$$

During $T_{OFF} = 0.8T_s$,

$$v_o(t) = 0$$

(b) The motor current waveform is shown below.

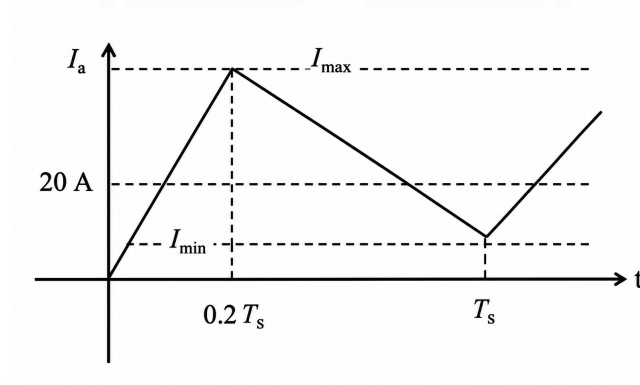


Fig. Solution Q3.91.1

The current increases linearly during T_{ON} and decreases linearly during T_{OFF} . The average current is

$$I_a = 20 \text{ A.}$$

(c) The duty cycle is

$$D = \frac{T_{ON}}{T_s} = 0.2$$

The average output voltage is

$$V_{avg} = DV_{in} = 0.2 \times 400 = 80 \text{ V}$$

During the ON interval, the voltage across the armature inductance is

$$V_L = V_{in} - E = 400 - 80 = 320 \text{ V}$$

Using

$$V_L = L \frac{\Delta i_L}{T_{ON}}$$
$$L \frac{\Delta i_L}{T_{ON}} = 320$$

Given,

$$L = 10 \text{ mH} = 10^{-2} \text{ H}, \quad T_{ON} = 0.2T_s$$

Therefore,

$$\Delta i_L = \frac{320 \times 0.2T_s}{10^{-2}} = 6400T_s$$

Hence, the peak-to-peak current ripple is

$$\boxed{6400T_s}$$



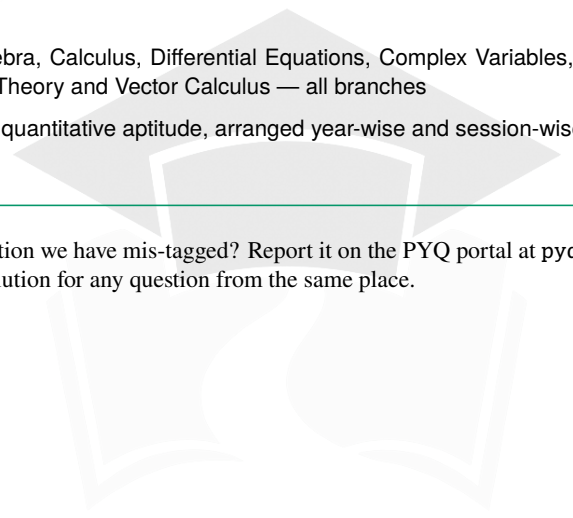
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